

- **Lab Title:** Nmap Mastery and Vulnerability Mapping
- **Assignment Title:** Nmap Mastery and Vulnerability Mapping
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- **Student ID:** 2025/FWSD/11509
- **Course Name:** BVWS102 – Web Application Security Essentials
- **Instructor Name:** Aminu Idris
- **Date of Submission:** 23 JAN, 2026
- **Version:** Version 1.3.0

LAB OVERVIEW/OBJECTIVE

This lab assignment is designed to provide you with hands-on, practical experience in utilizing Nmap (Network Mapper) for comprehensive network reconnaissance and vulnerability mapping, a critical skill for any cybersecurity professional. You will apply the theoretical knowledge gained in the "Nmap: Basic to Advanced Network Scanning" module to audit two purposefully vulnerable systems: Metasploitable and OWASP Broken Web Applications (BWA)

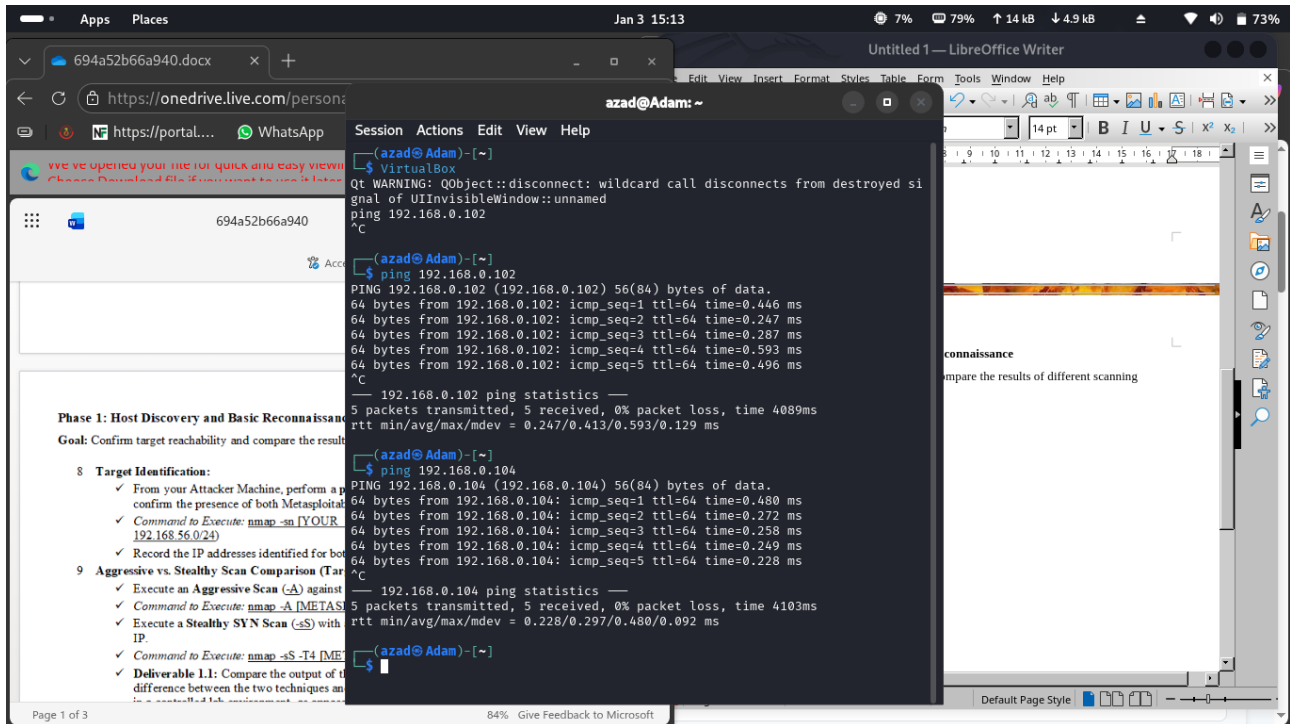
Required Components (Refer to your course materials for setup guides):

- 1 Virtualization Software: VirtualBox (Recommended) or VMware.
- 2 Attacker Machine: Kali Linux or Parrot OS (Pre-installed with Nmap).
- 3 Target 1 (Network Services): Metasploitable 2 or 3 VM.
- 4 Target 2 (Web Applications): OWASP Broken Web Applications (BWA) VM

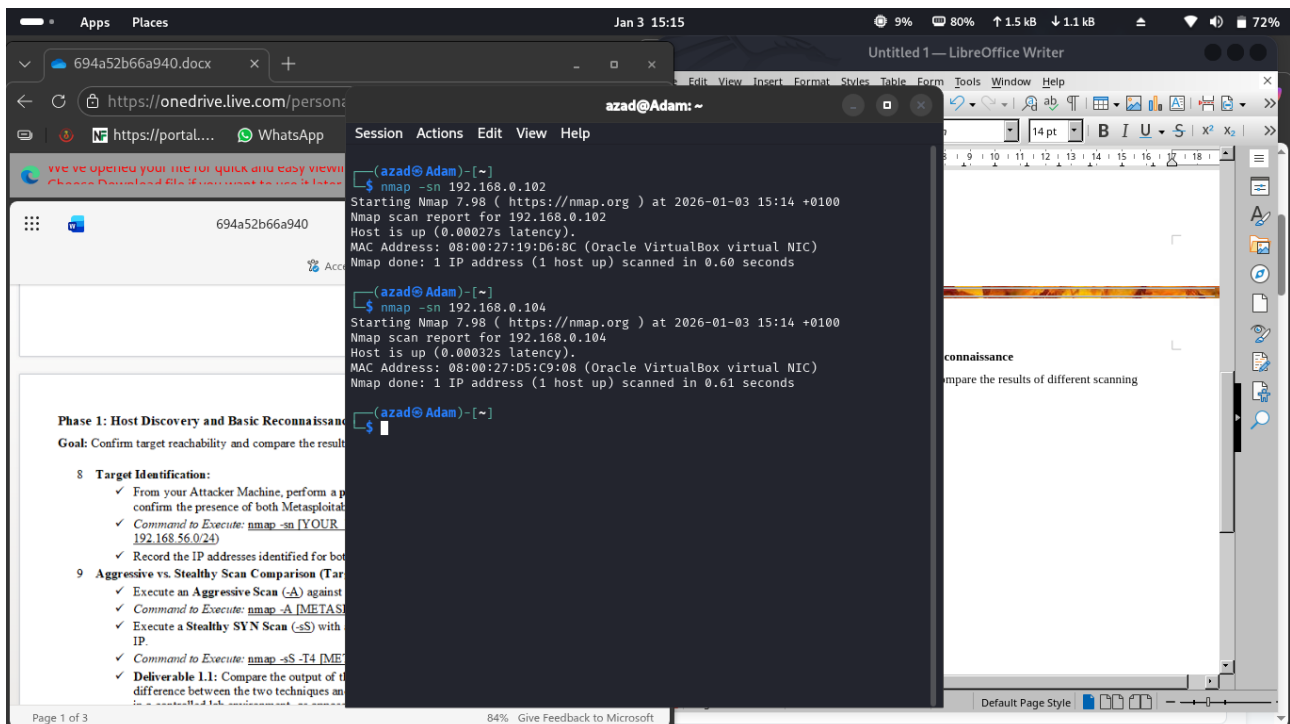
Phase 1: Host Discovery and Basic Reconnaissance (Target: Metasploitable):

Goal: Confirm target reachability and compare the results of different scanning methodologies.

I ping the both IP address to confirm perfect communication in both addresses



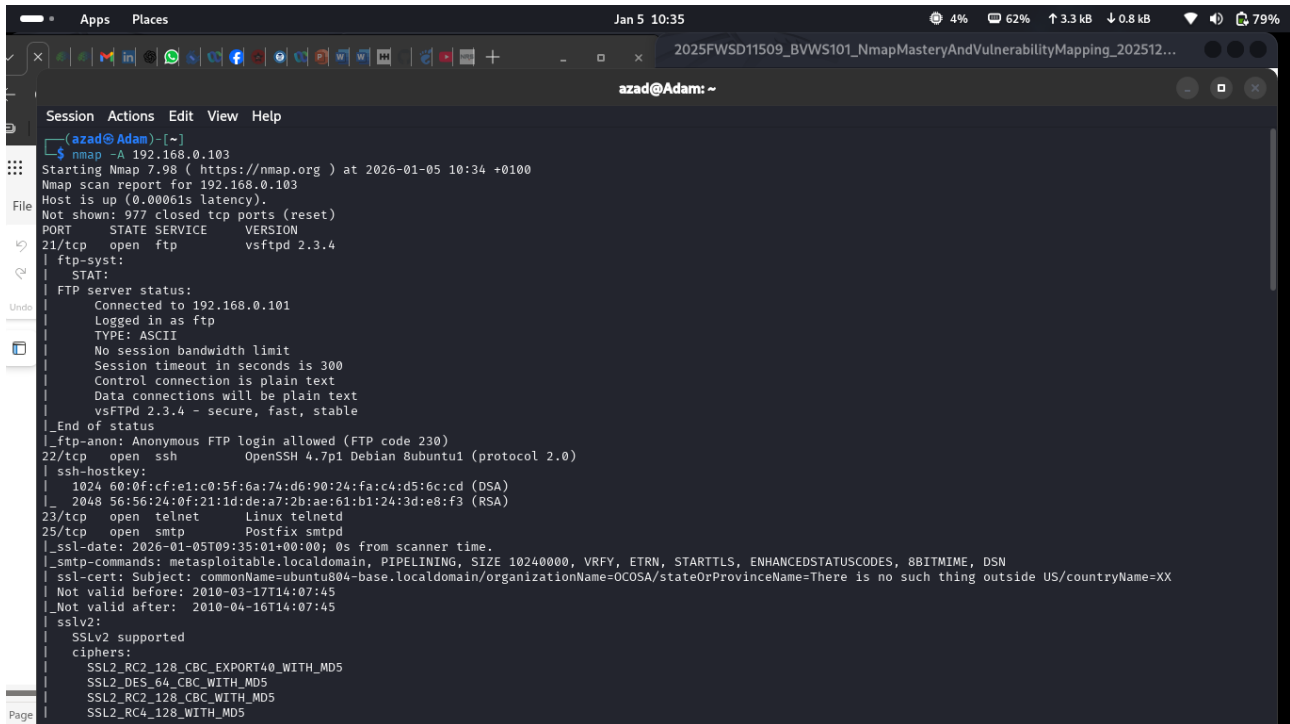
To confirm everything I make a normal scan using the command nmap -sn [IP]



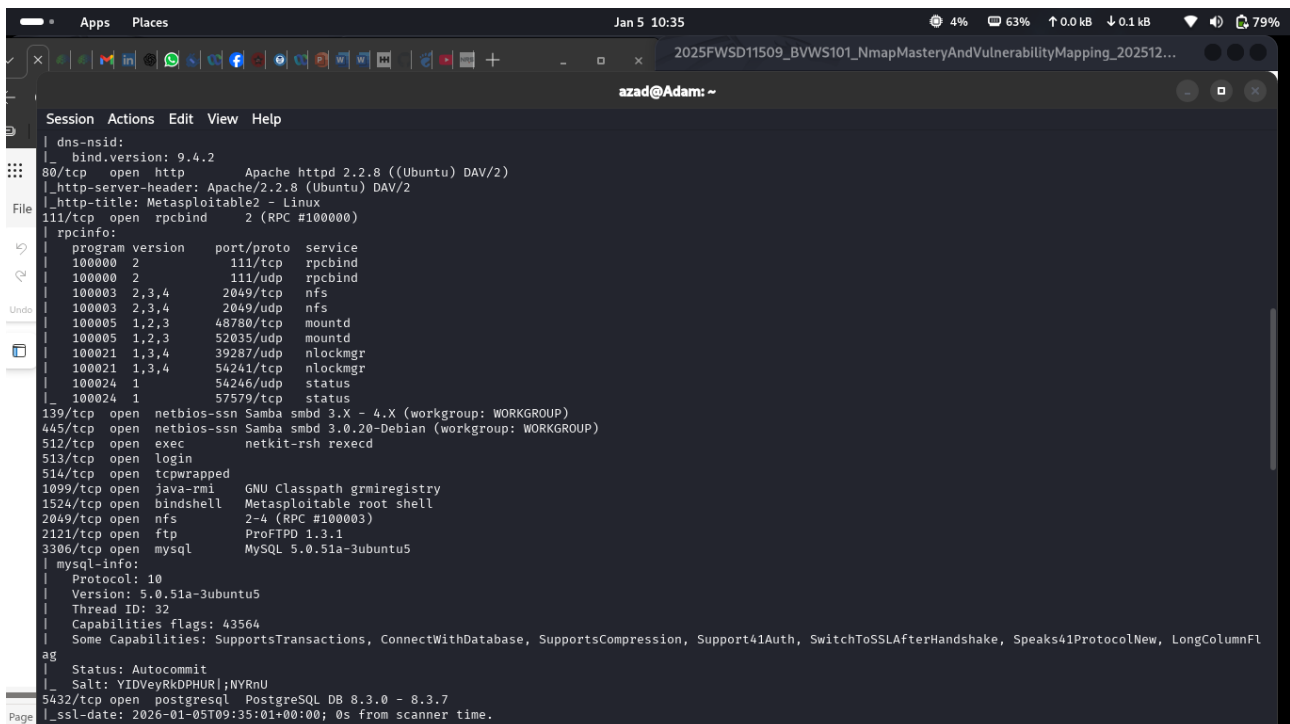
This is the part where I perform an aggressive scan

In this part I got all the necessary details I need completely from the target this get-out all the needful information in the target

It's better to use aggressive scanning in an isolated environment because you will get all the information you need and flag can be hidden inside in CTF challenge

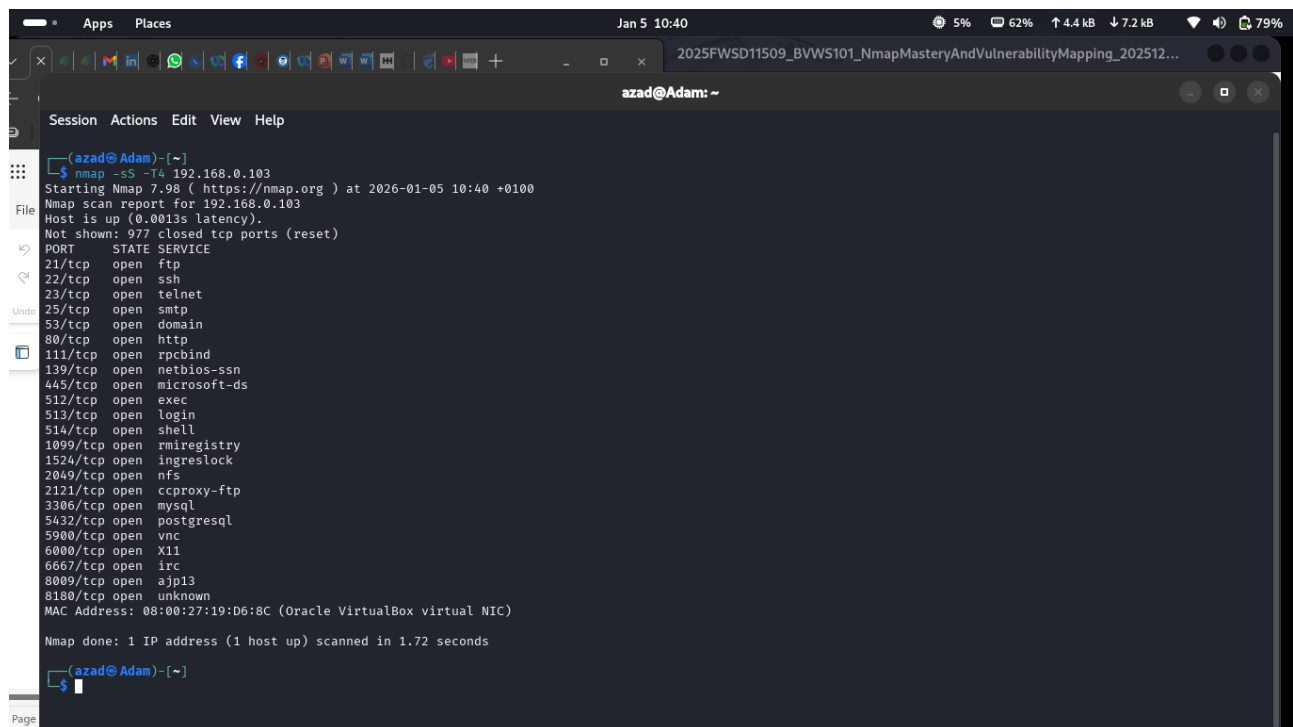


```
azad@Adam: ~  
$ nmap -A 192.168.0.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 10:34 +0100  
Nmap scan report for 192.168.0.103  
Host is up (0.00061s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
ftp-syst:  
STAT:  
FTP server status:  
Connected to 192.168.0.101  
Logged in as ftp  
TYPE: ASCII  
No session bandwidth limit  
Session timeout in seconds is 300  
Control connection is plain text  
Data connections will be plain text  
vsFTPd 2.3.4 - secure, fast, stable  
_End of status  
_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
ssh-hostkey:  
1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
_ssl-date: 2026-01-05T09:35:01+00:00; 0s from scanner time.  
_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN  
_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
Not valid before: 2010-03-17T14:07:45  
Not valid after: 2010-04-16T14:07:45  
_ssl-v2:  
SSLv2 supported  
ciphers:  
SSL2_RC2_128_CBC_EXPORT40_WITH_MD5  
SSL2_DES_64_CBC_WITH_MD5  
SSL2_RC2_128_CBC_WITH_MD5  
SSL2_RC4_128_WITH_MD5
```



```
azad@Adam: ~  
$ nmap -A 192.168.0.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 10:34 +0100  
Nmap scan report for 192.168.0.103  
Host is up (0.00061s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
_http-title: Metasploitable2 - Linux  
111/tcp    open  rpcbind      2 (RPC #100000)  
rpcinfo:  
program version port/proto service  
100000 2 111/tcp rpcbind  
100000 2 111/udp rpcbind  
100003 2,3,4 2049/tcp nfs  
100003 2,3,4 2049/udp nfs  
100005 1,2,3 48780/tcp mountd  
100005 1,2,3 52035/udp mountd  
100021 1,3,4 39287/udp nlockmgr  
100021 1,3,4 54241/tcp nlockmgr  
100024 1 54246/udp status  
100024 1 57579/tcp status  
139/tcp    open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp    open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)  
512/tcp    open  exec         netkit-rsh rexecd  
513/tcp    open  login        login  
514/tcp    open  tcpwrapped  
1099/tcp    open  java-rmi     GNU Classpath grmiregistry  
1524/tcp    open  bindshell    Metasploitable root shell  
2049/tcp    open  nfs          2-4 (RPC #100003)  
2121/tcp    open  ftp          ProFTPD 1.3.1  
3306/tcp    open  mysql        MySQL 5.0.51a-3ubuntu5  
mysql-info:  
Protocol: 10  
Version: 5.0.51a-3ubuntu5  
Thread ID: 32  
Capabilities: 43564  
Some Capabilities: SupportsTransactions, ConnectWithDatabase, SupportsCompression, Support41Auth, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, LongColumnFl  
ag  
Status: Autocommit  
_Salt: YIDVeyRkDPHURl;NVRnU  
5432/tcp    open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
_ssl-date: 2026-01-05T09:35:01+00:00; 0s from scanner time.
```

This is the part of stealthy SYN scan where I got few information on the port and it don't give the complete information. Some detail are hidden



The screenshot shows a terminal window titled "azad@Adam: ~" with a menu bar (Session, Actions, Edit, View, Help). The terminal displays the output of an Nmap SYN scan command: `azad@Adam:~$ nmap -sS -T4 192.168.0.103`. The output indicates the scan was performed on 2026-01-05 at 10:40 +0100. It reports that the host is up with a latency of 0.0013s and that 977 closed TCP ports were reset. A list of open ports is shown, each with its port number, protocol (tcp), state (open), and service. The services include ftp, ssh, telnet, smtp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmiregistry, ingreslock, nfs, ccproxy-ftp, mysql, postgresql, vnc, X11, irc, ajp13, and an unknown service on port 8180. The MAC address is listed as 08:00:27:19:D6:8C (Oracle VirtualBox virtual NIC). The scan completed in 1.72 seconds, having scanned 1 IP address with 1 host up.

```
(azad@ Adam)~$ nmap -sS -T4 192.168.0.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 10:40 +0100
Nmap scan report for 192.168.0.103
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:19:D6:8C (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.72 seconds
(azad@ Adam)~$
```

Phase 2: Detailed Service Enumeration and Version Detection

Goal: Precisely identify the services and their versions running on the targets to inform potential exploitation paths.

Comprehensive Service Versioning (Target: Metasploitable):

I perform a scan to know the specific version of all the port in the target sever but here are the exact version numbers for the following services:

- FTP (Port 21)
- Samba (Ports 139 and 445)
- SSH (Port 22)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

21/tcp	open	ftp	vsftpd 2.3.4
--------	------	-----	--------------

22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
--------	------	-----	--

139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
---------	------	-------------	---

445/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
---------	------	-------------	---

```
azad@Adam: ~  
$ nmap -sV 192.168.0.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 11:18 +0100  
Nmap scan report for 192.168.0.103  
Host is up (0.00095s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login          
514/tcp   open  tcpwrapped    
1099/tcp  open  java-rmi     GNU Classpath gmicregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 08:00:27:19:D6:8C (Oracle VirtualBox virtual NIC)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.36 seconds
```

Web Service Focus (Target: OWASP BWA):

In this case I run a particular scan in only port 80 and 443 in **OWASPBWA IP**

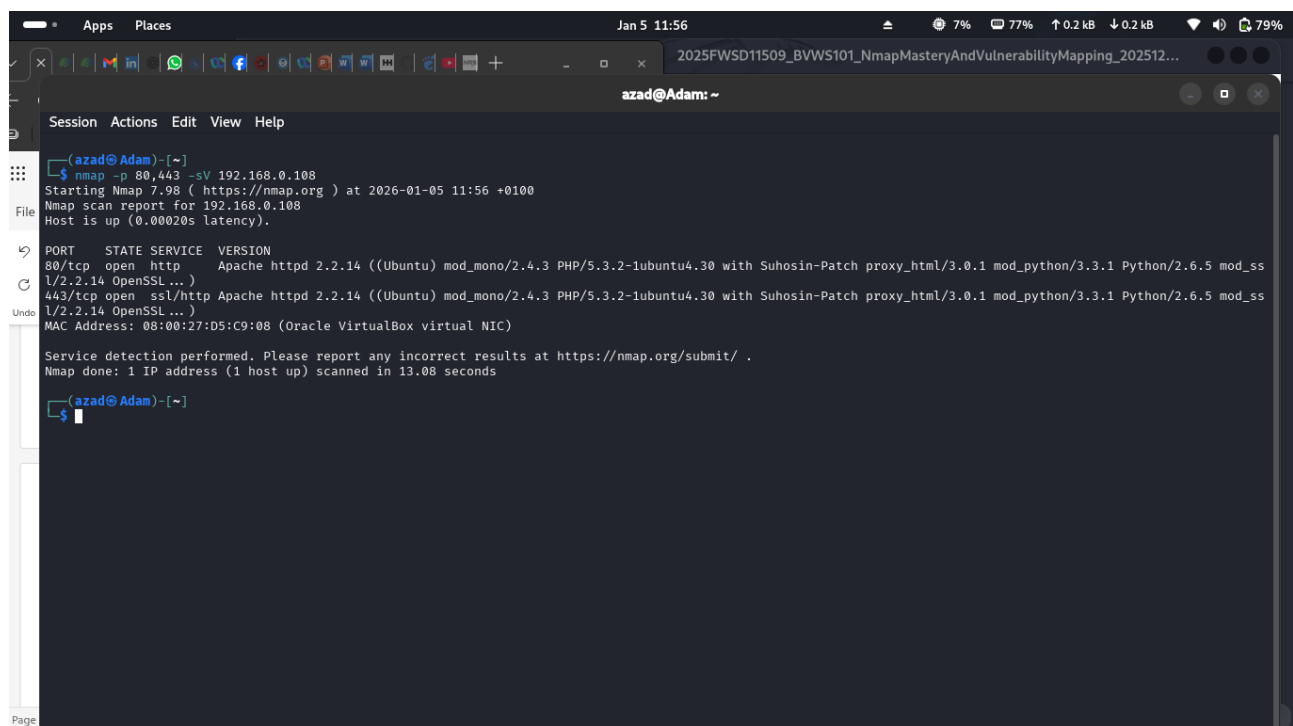
PORT STATE SERVICE VERSION

* 80/tcp open http Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL...)

* 443/tcp open ssl/http Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL...)

MAC Address: 08:00:27:D5:C9:08 (Oracle VirtualBox virtual NIC)

This figure out all the useful data regarding port 80 and 443



```
azad@Adam: ~  
$ nmap -p 80,443 -sV 192.168.0.108  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 11:56 +0100  
Nmap scan report for 192.168.0.108  
Host is up (0.00020s latency).  
  
PORT      STATE SERVICE VERSION  
80/tcp    open  http   Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL...)  
443/tcp    open  ssl/http Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL...)  
MAC Address: 08:00:27:D5:C9:08 (Oracle VirtualBox virtual NIC)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.08 seconds  
  
azad@Adam: ~$
```

Phase 3: Vulnerability Scanning with the Nmap Scripting Engine (NSE)

Goal: Automate the detection of common vulnerabilities using Nmap's powerful scripting engine.

Network Vulnerability Check (Target: Metasploitable):

These are some possible vulnerabilities that we can have in the metasploitable which all allow anonymous login and unauthenticated services using the command **nmap -sC -sV --script vuln 192.168.0.103**

21/tcp open ftp vsftpd 2.3.4

| ftp-vsftpd-backdoor:

| VULNERABLE:

| vsFTPD version 2.3.4 backdoor

| State: VULNERABLE (Exploitable)

| IDs: BID:48539 CVE:CVE-2011-2523

| vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.

| Disclosure date: 2011-07-03

25/tcp open smtp Postfix smtpd

| smtp-vuln-cve2010-4344:

|_ The SMTP server is not Exim: NOT VULNERABLE

| ssl-poodle:

| VULNERABLE:

| SSL POODLE information leak

| State: VULNERABLE

| IDs: BID:70574 CVE:CVE-2014-3566

| The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other products, uses nondeterministic CBC padding, which makes it easier for man-in-the-middle attackers to obtain cleartext data via a padding-oracle attack, aka the "POODLE" issue.

5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

| ssl-dh-params:

| VULNERABLE:

| Diffie-Hellman Key Exchange Insufficient Group Strength

| State: VULNERABLE

| Transport Layer Security (TLS) services that use Diffie-Hellman groups of insufficient strength, especially those using one of a few commonly shared groups, may be susceptible to passive eavesdropping attacks.

| Check results:

ssl-ccs-injection:

| VULNERABLE:

| SSL/TLS MITM vulnerability (CCS Injection)

| State: VULNERABLE

| Risk factor: High

| OpenSSL before 0.9.8za, 1.0.0 before 1.0.0m, and 1.0.1 before 1.0.1h does not properly restrict processing of ChangeCipherSpec messages, which allows man-in-the-middle attackers to trigger use of a zero length master key in certain OpenSSL-to-OpenSSL communications, and consequently hijack sessions or obtain sensitive information, via a crafted TLS handshake, aka the "CCS Injection" vulnerability.

ssl-poodle:

| VULNERABLE:

| SSL POODLE information leak

| State: VULNERABLE

| IDs: BID:70574 CVE:CVE-2014-3566

| The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other

products, uses nondeterministic CBC padding, which makes it easier for man-in-the-middle attackers to obtain cleartext data via a padding-oracle attack, aka the "POODLE" issue.

```
azad@Adam: ~  
$ nmap -sC -sV --script vuln 192.168.0.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 12:25 +0100  
Nmap scan report for 192.168.0.103  
Host is up (0.00075s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ ftp-vsftpd-backdoor:  
|_ VULNERABLE:  
|_ vsftpd version 2.3.4 backdoor  
|_   State: VULNERABLE (Exploitable)  
|_   IDs:   BID:48539  CVE:CVE-2011-2523  
|_   vsftpd version 2.3.4 backdoor, this was reported on 2011-07-04.  
|_   Disclosure date: 2011-07-03  
|_   Exploit results:  
|_     Shell command: id  
|_     Results: uid=0(root) gid=0(root)  
|_   References:  
|_     https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb  
|_     http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html  
|_     https://www.securityfocus.com/bid/48539  
|_     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523  
|_   vulners:  
|_     vsftpd 2.3.4:  
|_       PACKETSTORM:162145  10.0  https://vulners.com/packetstorm/PACKETSTORM:162145  *EXPLOIT*  
|_       EDB-ID:49757  10.0  https://vulners.com/exploitdb/EDB-ID:49757  *EXPLOIT*  
|_       E980AEB8-5138-50BF-8922-2D87E3C046DD  10.0  https://vulners.com/githubexploit/E980AEB8-5138-50BF-8922-2D87E3C046DD  *EXPLOIT*  
|_       CVE-2011-2523  10.0  https://vulners.com/cve/CVE-2011-2523  
|_       CNVD-2020-46837  10.0  https://vulners.com/cnvd/CNVD-2020-46837  
|_       CC3F6C15-182F-53F6-A5CC-812037F1F047  10.0  https://vulners.com/githubexploit/CC3F6C15-182F-53F6-A5CC-812037F1F047  *EXPLOIT*  
|_       A4185EAD-1A4C-56A6-97C6-1C58A1CF1E38  10.0  https://vulners.com/githubexploit/A4185EAD-1A4C-56A6-97C6-1C58A1CF1E38  *EXPLOIT*  
|_       817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3  10.0  https://vulners.com/githubexploit/817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3  *EXPLOIT*  
|_       63A5C9A7-C241-5E83-9EE6-ABAB44BD0270  10.0  https://vulners.com/githubexploit/63A5C9A7-C241-5E83-9EE6-ABAB44BD0270  *EXPLOIT*  
|_       5F48CEDE-77DF-5D54-851A-0AE8B76458D9  10.0  https://vulners.com/githubexploit/5F48CEDE-77DF-5D54-851A-0AE8B76458D9  *EXPLOIT*  
|_       50580586-73C4-5097-81CA-546D6591DF44  10.0  https://vulners.com/githubexploit/50580586-73C4-5097-81CA-546D6591DF44  *EXPLOIT*  
|_       1337DAY-ID-36095  9.8  https://vulners.com/zdt/1337DAY-ID-36095  *EXPLOIT*  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
```

```
azad@Adam: ~  
$ nmap -sC -sV --script vuln 192.168.0.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 12:25 +0100  
Nmap scan report for 192.168.0.103  
Host is up (0.00075s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ ftp-vsftpd-backdoor:  
|_ VULNERABLE:  
|_ vsftpd version 2.3.4 backdoor  
|_   State: VULNERABLE (Exploitable)  
|_   IDs:   BID:48539  CVE:CVE-2011-2523  
|_   vsftpd version 2.3.4 backdoor, this was reported on 2011-07-04.  
|_   Disclosure date: 2011-07-03  
|_   Exploit results:  
|_     Shell command: id  
|_     Results: uid=0(root) gid=0(root)  
|_   References:  
|_     https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb  
|_     http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html  
|_     https://www.securityfocus.com/bid/48539  
|_     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523  
|_   vulners:  
|_     vsftpd 2.3.4:  
|_       PACKETSTORM:162145  10.0  https://vulners.com/packetstorm/PACKETSTORM:162145  *EXPLOIT*  
|_       EDB-ID:49757  10.0  https://vulners.com/exploitdb/EDB-ID:49757  *EXPLOIT*  
|_       E980AEB8-5138-50BF-8922-2D87E3C046DD  10.0  https://vulners.com/githubexploit/E980AEB8-5138-50BF-8922-2D87E3C046DD  *EXPLOIT*  
|_       CVE-2011-2523  10.0  https://vulners.com/cve/CVE-2011-2523  
|_       CNVD-2020-46837  10.0  https://vulners.com/cnvd/CNVD-2020-46837  
|_       CC3F6C15-182F-53F6-A5CC-812037F1F047  10.0  https://vulners.com/githubexploit/CC3F6C15-182F-53F6-A5CC-812037F1F047  *EXPLOIT*  
|_       A4185EAD-1A4C-56A6-97C6-1C58A1CF1E38  10.0  https://vulners.com/githubexploit/A4185EAD-1A4C-56A6-97C6-1C58A1CF1E38  *EXPLOIT*  
|_       817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3  10.0  https://vulners.com/githubexploit/817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3  *EXPLOIT*  
|_       63A5C9A7-C241-5E83-9EE6-ABAB44BD0270  10.0  https://vulners.com/githubexploit/63A5C9A7-C241-5E83-9EE6-ABAB44BD0270  *EXPLOIT*  
|_       5F48CEDE-77DF-5D54-851A-0AE8B76458D9  10.0  https://vulners.com/githubexploit/5F48CEDE-77DF-5D54-851A-0AE8B76458D9  *EXPLOIT*  
|_       50580586-73C4-5097-81CA-546D6591DF44  10.0  https://vulners.com/githubexploit/50580586-73C4-5097-81CA-546D6591DF44  *EXPLOIT*  
|_       1337DAY-ID-36095  9.8  https://vulners.com/zdt/1337DAY-ID-36095  *EXPLOIT*  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
|_ smtp-vuln-cve2010-4344:  
|_   The SMTP server is not Exim: NOT VULNERABLE  
|_   ssl-poodle:  
|_     VULNERABLE:  
|_     SSL POODLE information leak  
|_     State: VULNERABLE  
|_     IDs:   BID:70574  CVE:CVE-2014-3566  
|_     The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other products, uses nondeterministic CBC padding, which makes it easier for man-in-the-middle attackers to obtain cleartext data via a padding-oracle attack, aka the "POODLE" issue.  
|_     Disclosure date: 2014-10-14  
|_     Check results:  
|_       TLS_RSA_WITH_AES_128_CBC_SHA  
|_     References:  
|_       https://www.imperialviolet.org/2014/10/14/poodle.html  
|_       https://www.openssl.org/~bodo/ssl-poodle.pdf  
|_       https://www.securityfocus.com/bid/70574  
|_       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-3566  
|_     ssl-dh-params:  
|_       VULNERABLE:  
|_       Anonymous Diffie-Hellman Key Exchange MitM Vulnerability  
|_       State: VULNERABLE  
|_       Transport Layer Security (TLS) services that use anonymous Diffie-Hellman key exchange only provide protection against passive eavesdropping, and are vulnerable to active man-in-the-middle attacks which could completely compromise the confidentiality and integrity
```

Web Application Enumeration (Target: OWASP BWA):

In this part I scan the port 80 and got some hidden directories and application in the owaspbwa and this show or give the high possibility of intrusion and give hackers' chance to div in to the system directly

PORT STATE SERVICE

80/tcp open http

|_http-title: owaspbwa OWASP Broken Web Applications

| http-enum:

| /wordpress/: Blog

| /test/: Test page

| /mono/: Mono

| /crossdomain.xml: Adobe Flash crossdomain policy

| /phpmyadmin/: phpMyAdmin

| /wordpress/wp-login.php: Wordpress login page.

| /cgi-bin/: Potentially interesting folder w/ directory listing

| /icons/: Potentially interesting folder w/ directory listing

|_ /images/: Potentially interesting folder w/ directory listing

```
azad@Adam: ~  
$ nmap -p 80 --script http-enum,http-title 192.168.0.108  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-05 12:57 +0100  
Nmap scan report for 192.168.0.108  
Host is up (0.00028s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
_http-title: owaspbwa OWASP Broken Web Applications  
_http-enum:  
| /wordpress/: Blog  
| /test/: Test page  
| /mono/: Mono  
| /crossdomain.xml: Adobe Flash crossdomain policy  
| /phpmyadmin/: phpMyAdmin  
| /wordpress/wp-login.php: Wordpress login page.  
| /cgi-bin/: Potentially interesting folder w/ directory listing  
| /icons/: Potentially interesting folder w/ directory listing  
| /images/: Potentially interesting folder w/ directory listing  
_MAC Address: 08:00:27:D5:C9:08 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 7.28 seconds  
  
azad@Adam: ~  
$
```

CONCLUSSION

At the end of this lab i got to know the important of the tool nmap to the real world action and I also learn the following:

- Set up a secure and isolated virtual lab environment using VirtualBox.
- Perform effective host discovery and target identification.
- Differentiate and execute various Nmap port scanning techniques (SYN, TCP Connect, UDP).
- Accurately detect operating systems and perform detailed service version enumeration.
- Leverage the Nmap Scripting Engine (NSE) for automated vulnerability assessment and web enumeration.